

datasets for training, and `X_val` and `y_val` are for validation. The batch size that we had fixed earlier to 32 tells the data generator to take only the specified batch of 32 images at a time.

2. Now we train our neural network by calling `.fit()` method on the model and passing some parameters. The first parameter is the training set `Image-DataGenerator` object or the `train_generator`.
3. Next, we specify the number of steps per epoch that tells our model how many images we want to process before making a gradient update in our loss function.

Since our batch size is 32 and there are 3200 images, it means there will be $3200/32$, which is 100 steps. This means that we are going to make 100 gradient update to our model in one pass through the entire training set.

An epoch is a full cycle through the entire training set. In our model, an epoch is reached when we make 100 gradient updates as calculated above using our steps per epoch method. We specify the Epochs to be 64 that means we want to go over our training data 64 times and each time we will make 100 times gradient updates.

4. No pass in our validation data generator. Set the step size here too. It is a general practise to set the same steps as that of the training data. Hence we are also going to do the same.

Just 64 epochs, our model got an accuracy of roughly 80%. Increasing the number of epochs and tuning some hyper-parameters like batch size, learning rates, input size, network depth, backpropagation algorithms, optimisers etc. may help improve the accuracy of the model. Fine-tuning the hyper - parameters may increase the accuracy of the model upto 90% but not beyond that. Hence, we use the concept of transfer learning to increase the accuracy beyond 90% that will be elaborated further in this article.

```
100/100 [*****] - 19s 194ms/step - loss: 0.6415 - acc: 0.7922 - val_loss: 0.3764 - val_acc: 0.8858
Epoch 52/64
100/100 [*****] - 20s 193ms/step - loss: 0.6478 - acc: 0.7889 - val_loss: 0.4476 - val_acc: 0.8823
Epoch 53/64
100/100 [*****] - 19s 194ms/step - loss: 0.6382 - acc: 0.7944 - val_loss: 0.5188 - val_acc: 0.7708
Epoch 54/64
100/100 [*****] - 20s 193ms/step - loss: 0.6421 - acc: 0.8076 - val_loss: 0.6173 - val_acc: 0.8143
Epoch 55/64
100/100 [*****] - 19s 193ms/step - loss: 0.6427 - acc: 0.7979 - val_loss: 0.6172 - val_acc: 0.8213
Epoch 56/64

100/100 [*****] - 19s 194ms/step - loss: 0.6714 - acc: 0.8004 - val_loss: 0.4538 - val_acc: 0.8863
Epoch 62/64
100/100 [*****] - 20s 201ms/step - loss: 0.4362 - acc: 0.7957 - val_loss: 0.4013 - val_acc: 0.7358
Epoch 63/64
100/100 [*****] - 19s 194ms/step - loss: 0.4228 - acc: 0.7967 - val_loss: 0.4418 - val_acc: 0.8863
Epoch 64/64
100/100 [*****] - 20s 197ms/step - loss: 0.4268 - acc: 0.8083 - val_loss: 0.4272 - val_acc: 0.8823
```

Figure : Training of model in progress